MAT3032 ANSWERS TO EXERCISES 1.2

1. $\mathrm{A}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ so $\mathrm{A}^4 = (\mathrm{A}^2)^2 = \mathrm{I}$. $\mathrm{B}^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = \mathrm{A}^2$.

   $\mathrm{ABA} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \mathrm{B}$.

   Also $\mathrm{A}^3 = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$, $\mathrm{AB} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$, $\mathrm{A}^2\mathrm{B} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $\mathrm{A}^3\mathrm{B} = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}$
   so $\{\mathrm{I}, \mathrm{A}, \mathrm{A}^2, \mathrm{A}^3, \mathrm{B}, \mathrm{AB}, \mathrm{A}^2\mathrm{B}, \mathrm{A}^3\mathrm{B}\}$ is a group of eight matrices with the presentation $\langle \mathrm{A}, \mathrm{B} : \mathrm{A}^4 = \mathrm{I}, \mathrm{B}^2 = \mathrm{A}^2, \mathrm{ABA} = \mathrm{B}\rangle$. Thus it is isomorphic to Q_4.

2. The group is $C_3 \times C_2$, which has order 6. This is isomorphic to C_6, so it has the presentation $\langle g \ : \ g^6 = e\rangle$ where g is any generator.

   Taking $g = ab$, the powers of g are ab, a^2, b, a, a^2b, e respectively, so ab is a generator. OR: taking $g = a^2b$, the powers of g are a^2b, a, b, a^2, ab, e respectively, so a^2b is a generator.

3. $\sigma = (1\ \ 2\ \ 3\ \ 4\ \ 5)$ is a rotation through $\dfrac{2\pi}{5}$ about the centre of the polygon.

   $\tau = (2\ \ 5)(3\ \ 4)$ is a reflection in the line of symmetry through vertex 1.

   The elements are ι, of order 1; $\sigma, \sigma^2, \sigma^3, \sigma^4$ all of order 5, and $\tau, \sigma\tau, \sigma^2\tau, \sigma^3\tau, \sigma^4\tau$ all of order 2.

   The subgroups of D_5 are $\{\iota\}$ (trivial); $\{\iota, \tau\}$, $\{\iota, \sigma\tau\}$, $\{\iota, \sigma^2\tau\}$, $\{\iota, \sigma^3\tau\}$, $\{\iota, \sigma^4\tau\}$, $\{\iota, \sigma, \sigma^2, \sigma^3, \sigma^4\}$ and D_5 itself.

4. If $|G| = 6$ then by Lagrange's theorem, all elements have order 1, 2, 3 or 6. Only the identity can have order 1. If there were an element of order 6 then G would be cyclic, hence abelian.

   By Exercises 1.1 Question 8, G has at least one element of order 2. If *all* non-identity elements have order 2, G is abelian by Exercises 1.1 Q. 2 (b).

   Thus G has an element a of order 3 and an element b of order 2. As $\langle a\rangle$ and $\langle b\rangle$ have coprime orders, $b \notin \langle a\rangle$. Then b, ab, a^2b are distinct from one another and from e, a, a^2, so $G = \{e, a, a^2, b, ab, a^2b\}$.

   $ba \notin \langle a\rangle$, so $ba \in \{b, ab, a^2b\}$. As $a \neq e, ba \neq b$, and as G is non-abelian, $ba \neq ab$. Thus $ba = a^2b$, so $aba = a^3b = b$.

   Hence G has the presentation $\langle a, b \ : \ a^3 = b^2 = e, aba = b\rangle$, so $G \cong D_3$.

5. e has order 1, a^2 has order 2. All other elements have order 4.

   The subgroups are $\{e\}$, $\{e, a^2\}$, $\{e, a, a^2, a^3\}$, $\{e, a^2, b, a^2b\}$, $\{e, a^2, ab, a^3b\}$ and Q_4 itself. (Note that any group of order 4 must have an element of order 2, and a^2 is the only such element.)

   Since there is only one self-inverse element, the 4-element subgroups are not isomorphic to V. Hence all the proper subgroups are cyclic: $\{e, a^2\}$ is generated by a^2, $\{e, a, a^2, a^3\}$ is generated by a (or a^3), $\{e, b, a^2, a^2b\}$ is generated by b (or a^2b), $\{e, ab, a^2, a^3b\}$ is generated by ab (or a^3b).